Discussion Point:

Which network topologies have you used? You might wish to defer this discussion until you have read about network devices later in this chapter.

In a situation where several LANs are connected, they can have different topologies or supporting technologies. This collection of LANs then becomes a **hybrid network**. A special connecting device is needed to ensure that the hybrid network is fully functional. It is often an advantage to be able to connect a new topology LAN to existing LANs where it is not sensible or not possible to use the existing topology for the new LAN. An example is when a wired LAN is already installed but a new wireless LAN is to be connected to it.

KEY TERMS

Hybrid network: a collection of connected LANs where some of them have different topologies or supporting technologies

Cable: a transmission using copper wire or fibre-optic

Bandwidth: a measure of the amount of data that can be transmitted per second

2.03 Transmission media

Cable

A network **cable** can be twisted pair, coaxial or fibre-optic. The twisted pair and coaxial cables both use copper for the transmission medium. In discussing suitability for a given application there are a number of factors to consider. One of these is the cost of the cable and connecting devices. Another is the best **bandwidth** that can be achieved. The bandwidth governs the possible data transmission rate. There are then two factors that can cause poor performance: the likelihood of interference affecting transmitted signals and the extent of attenuation (deterioration of the signal) when high frequencies are transmitted. These factors will dictate whether repeaters or amplifiers are needed in transmission lines and how many will be needed. Table 2.01 shows some comparisons of the different cable types.

	Twisted pair	Coaxial	Fibre-optic
Cost	Lowest	Higher	Highest
Bandwidth or data rate	Lowest	Higher	Much higher
Attenuation at high frequency	Affected	Most affected	Least affected
Interference	Worst affected	Less affected	Least affected
Need for repeaters	More often	More often	Less often

Table 2.01 Comparisons between cable types

You need to understand that for each of the three types of cabling there are defined standards for different grades of cable which must be considered when you decide which type of cable to use. Fibre-optic cable performs best but costs more than the other kinds. For a new installation the improved performance of fibre-optic cable is likely to be the factor that governs your choice. However, where copper cable is already installed the cost of replacement by fibre-optic cable may not be justified.

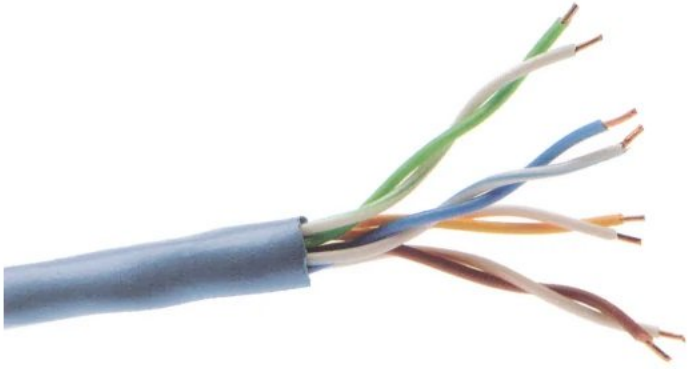


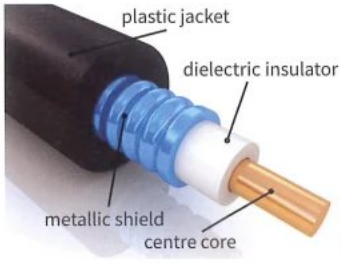
Figure 2.05 One cable with four twisted pairs with differing twist rates to reduce interference

Currently, twisted pair cable is normally used to connect telephone handsets to telephone lines. This type of cable is illustrated in Figure 2.05. It is also the technology of choice for high-speed local area networks.

Question 2.01

Twisted pair cable can be shielded or unshielded. What are the options for this? How does shielding affect the use of the cable?

Coaxial cable is used extensively by cable television companies and in metropolitan area networks. It is not usually used for long-distance telephone cabling. Fibre-optic cable is the technology of choice for long-distance cabling. As shown in Figure 2.06, coaxial cable is not bundled but a fibre-optic cable contains many individual fibres.



a



b

Figure 2.06 (a) Coaxial cable and (b) a bundled fibre-optic cable

Wireless

The alternative to cable is **wireless** transmission. The three options here are radio, microwave or infrared. These are all examples of electromagnetic radiation; the only intrinsic difference between the three types is the frequency of the waves.

KEY TERM

Wireless: a transmission using radio, microwave or infrared

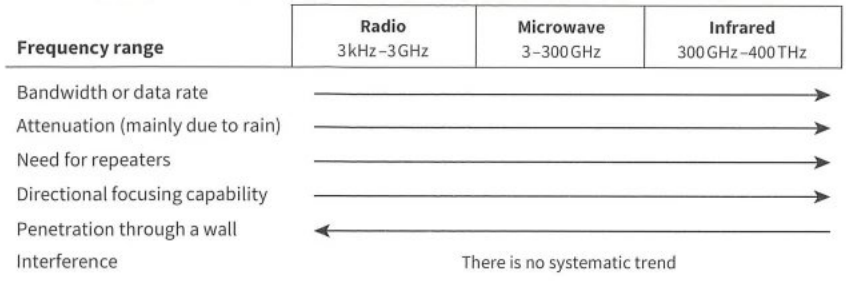


Figure 2.07 Frequency ranges and frequency dependency of factors affecting wireless transmission

When making a choice of which wireless option to use you need to consider all of the same factors that were discussed when comparing different kinds of cable. In addition, the ability of the radiation to transmit through a solid barrier is an important factor. Also, the extent to which the transmission can be focused in a specific direction needs to be considered. Figure 2.07 shows the approximate frequency ranges for the three types of radiation. The factors listed on the left increase in the direction of the arrows. The bandwidth increases through radio and microwave to infrared but the ability of the waves to penetrate solid objects is greatest for radio waves. Interference is not consistently affected by the frequency.

The increased attenuation for infrared transmission, which has the highest frequency, means that it is only suitable for indoor applications. The fact that it will not penetrate through a wall is then of benefit because the transmission cannot escape and cause unwanted interference elsewhere. For most applications, microwave transmission is the best option because it has a better bandwidth compared to that available using radio waves.

Comparing cable and wireless transmission

It is worth noting that cables are often referred to as 'guided media' and wireless as 'unguided media'. This is slightly misleading because only radio wave transmission fits the description of unguided. It is possible with microwaves or infrared to direct a transmission towards a particular receiver (as suggested in Figure 2.07).

There are other points to consider when we compare the relative advantages of transmission through a cable or wireless transmission.

- The use of certain wireless transmission frequencies is regulated by government agencies and so permission has to be obtained before wireless transmission is used.
- Outside these frequencies, no permission is needed to use the air for transmission but cables can only be laid in the ground with the permission of landowners.
- For global communications, the two competing technologies are: transmission through fibre-optic cables laid underground (or on the sea bed) and satellite transmission (discussed later in this section).
- Interference is much more significant for wireless transmission and its extent is dependent on which frequencies are being used for different applications.
- Repeaters are needed less often for wireless transmission.
- Mobile (cell) phones now dominate Internet use and for these, only wireless transmission is possible.
- For home or small office use, wired or wireless transmission is equally efficient; often, not having to install cables favours wireless connections for a small network.

Satellites are components of modern communication systems. Figure 2.08 shows the altitudes (distances above Earth) of three different types of satellite. The Van Allen belts are areas containing high levels of electrically charged particles, which interfere with satellites.

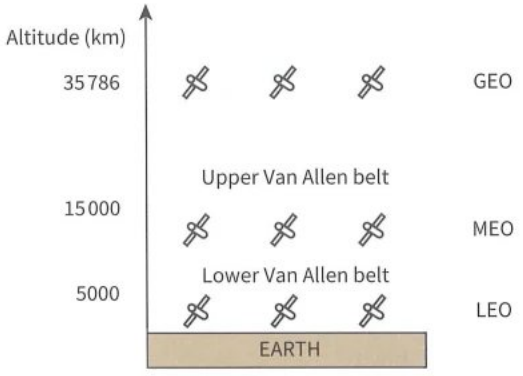


Figure 2.08 Satellite altitudes

The highest altitude satellites are in geostationary Earth orbit (GEO) over the equator and these are used to provide long-distance telephone and computer network communication. 'Geostationary' means that the satellite orbits at the same speed as the Earth spins, so from a point on the Earth the satellite always appears to be at the same point in the sky. Only three GEO satellites are needed for full global coverage. Closer to Earth are a group of medium-Earth-orbit (MEO) satellites some of which provide the global positioning system (GPS). Ten MEO satellites are needed for global coverage. Finally, low-Earth-orbit (LEO) satellites work in 'constellations' to supplement the mobile phone networks. Fifty LEO satellites are needed for full global coverage but currently there are several hundred LEO satellites in orbit.

A satellite can act as a component in a network and can directly connect with ground-based components. These ground-based components can be much further apart than in a network with no satellites. The disadvantage of satellites is that the greater transmission distance causes transmission delays, which can cause technical problems for the network.

TASK 2.01

Calculate the approximate time taken for a transmission from the surface of the Earth to a medium-Earth-orbit satellite. (Take the speed of light to be 300 000 km per second.)

The use of satellites in networks tends to be for specialised applications such as the Global Positioning System (GPS) or for Internet use in remote locations. At one stage, a lot of Internet communication was expected to make use of satellites, but the development of high-speed fibre-optic cabling at relatively low cost has reduced the need for satellites.

2.04 LAN hardware

Wired LANs

In the early years, coaxial cable was used for LANs. Nowadays, twisted pair cables are probably the most widely used networking connections, and fibre-optic cables are becoming more common. In a bus configuration the bus will consist of a series of sockets linked by cables. The ends of the bus have terminators attached that prevent signals from reflecting

Specifically, the web consists of an enormous collection of websites each having one or more web pages. The special feature of a web page is that it can contain hyperlinks which, when clicked, give direct and essentially immediate access to other web pages.

Cloud computing

Cloud computing is the provision of computing services usually via the Internet. An organisation may choose to establish its own **private cloud**. In this case there are three possible approaches:

- The organisation takes full responsibility for creating and managing the cloud installed on-site and connected to a private network
- The organisation outsources to a third-party the creation and management of an on-site installation connected to a private network
- The organisation outsources the creation and management of an Internet accessible system by a third-party.

The alternative is a **public cloud**. This is created, managed and owned by a third-party cloud service provider.

The services provided by a cloud are familiar ones provided by file servers and application servers. They are accessible via a browser and therefore accessible from any suitable device in any location. A public cloud can be accessed by an individual user or by an organisation. One major difference is the scale of the systems. The provision is established using large mainframe computers or server farms. The services provided can be characterised as being one of:

- infrastructure provision
- platform provision
- software provision

Many of the advantages to a cloud user arise from the fact that the cloud does not have the limitations that the systems already available have. For the infrastructure provision, the advantages include the better performance when running software and the increased storage capacity. For the platform provision, the cloud can offer facilities for software development and testing. For the software provision, the cloud will be able to run applications that require high performance systems. Alternatively, it could be that the costs to a company of buying and installing a software package themselves would be far too high. The other advantage is the familiar one with regard to outsourcing. The cloud user no longer needs technical expertise.

The disadvantages to a cloud user relate to the use of a public cloud. The cloud service provider has complete access to all of the data stored on the cloud. The cloud user cannot be sure that their data is not being shared with third-parties. This is a concern with regard to data privacy. The security of the data stored is an issue; the cloud service provider is being relied on to ensure data cannot be lost.



KEY TERMS

Private cloud: owned by and only accessed by an organisation

Public cloud: owned by a cloud service provider for general access